

2021 年数学(三) 真题解析

一、选择题

(1) 【答案】 (C).

【解】 由 $\int_0^{x^2} (e^{t^3} - 1) dt \sim \int_0^{x^2} t^3 dt = \frac{1}{4} x^8 (x \rightarrow 0)$ 得 $\int_0^{x^2} (e^{t^3} - 1) dt$ 为 x^7 的高阶无穷小, 应选(C).

(2) 【答案】 (D).

【解】 因为 $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1 = f(0)$, 所以 $f(x)$ 在 $x = 0$ 处连续;

由 $\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x} = \lim_{x \rightarrow 0} \frac{\frac{e^x - 1}{x} - 1}{x} = \lim_{x \rightarrow 0} \frac{e^x - 1 - x}{x^2} = \lim_{x \rightarrow 0} \frac{e^x - 1}{2x} = \frac{1}{2}$ 得 $f'(0) = \frac{1}{2} \neq 0$, 应选(D).

(3) 【答案】 (A).

【解】 因为 $f(x) = ax - b \ln x$ 有两个零点, 所以由罗尔定理, 存在 $c \in (0, +\infty)$, 使得 $f'(c) = a - \frac{b}{c} = 0$, 从而 $b = ac > 0$.

令 $f'(x) = a - \frac{b}{x} = 0$ 得 $x = \frac{b}{a}$,

因为 $f''(x) = \frac{b}{x^2} > 0$, 所以 $x = \frac{b}{a}$ 为函数 $f(x) = ax - b \ln x$ 的最小值点, 最小值为

$$f\left(\frac{b}{a}\right) = b - b \ln \frac{b}{a} = b \left(1 - \ln \frac{b}{a}\right),$$

又 $f(0+0) = +\infty, f(+\infty) = +\infty$,

所以 $f(x) = ax - b \ln x$ 有两个零点等价于 $b \left(1 - \ln \frac{b}{a}\right) < 0$, 即 $\frac{b}{a} > e$, 应选(A).

(4) 【答案】 (C).

【解】 $f(x+1, e^x) = x(x+1)^2$ 两边对 x 求导得

$$f'_1(x+1, e^x) + e^x f'_2(x+1, e^x) = (x+1)^2 + 2x(x+1),$$

取 $x=0$ 得 $f'_1(1, 1) + f'_2(1, 1) = 1$;

$f(x, x^2) = 2x^2 \ln x$ 两边对 x 求导得

$$f'_1(x, x^2) + 2x f'_2(x, x^2) = 4x \ln x + 2x,$$

取 $x=1$ 得 $f'_1(1, 1) + 2f'_2(1, 1) = 2$,

解得 $f'_1(1, 1) = 0, f'_2(1, 1) = 1$, 故 $df(1, 1) = dy$, 应选(C).

(5) 【答案】 (B).

【解】 将二次型展开得 $f(x_1, x_2, x_3) = 2x_2^2 + 2x_1x_2 + 2x_1x_3 + 2x_2x_3$,

其矩阵为 $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 0 \end{pmatrix}$, 由

$$|\lambda E - A| = \begin{vmatrix} \lambda & -1 & -1 \\ -1 & \lambda - 2 & -1 \\ -1 & -1 & \lambda \end{vmatrix} = (\lambda + 1) \begin{vmatrix} 1 & 0 & 0 \\ -1 & \lambda - 2 & -2 \\ -1 & -1 & \lambda - 1 \end{vmatrix} = (\lambda + 1)(\lambda^2 - 3\lambda) = 0$$

得 $\lambda_1 = -1, \lambda_2 = 0, \lambda_3 = 3$, 应选(B).

(6) 【答案】 (D).

【解】 因为 $A = (\alpha_1, \alpha_2, \alpha_3, \alpha_4)$ 为正交矩阵, 所以 $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ 两两正交且为单位向量, 因为 $\alpha_1, \alpha_2, \alpha_3$ 线性无关, 所以 $r(B) = 3 < 4$,

由 $B(\alpha_1 + \alpha_2 + \alpha_3) = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \beta$, 得 $\alpha_1 + \alpha_2 + \alpha_3$ 为方程组 $BX = \beta$ 的一个特解,

再由 $B\alpha_4 = 0$ 得 α_4 为 $BX = 0$ 的基础解系,

故方程组 $BX = \beta$ 的通解为 $X = \alpha_1 + \alpha_2 + \alpha_3 + k\alpha_4$, 应选(D).

(7) 【答案】 (C)

【解】 (A) 选项: $\begin{pmatrix} 1 & 0 & -1 \\ 2 & -1 & 1 \\ -1 & 2 & -5 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 2 & -1 & 0 \\ -1 & 2 & 0 \end{pmatrix}$, (A) 不对;

(B) 选项: $\begin{pmatrix} 1 & 0 & 0 \\ 2 & -1 & 0 \\ -3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 2 & -1 & 1 \\ -1 & 2 & -5 \end{pmatrix} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{pmatrix}$, (B) 不对;

(C) 选项: $\begin{pmatrix} 1 & 0 & 0 \\ 2 & -1 & 0 \\ -3 & 2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & -1 \\ 2 & -1 & 1 \\ -1 & 2 & -5 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{pmatrix}$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix},$$

应选(C).

(8) 【答案】 (D).

【解】 由 $P(A | B) = P(A)$ 得 $P(AB) = P(A)P(B)$, 即事件 A, B 独立,

于是 $P(A | \bar{B}) = \frac{P(A\bar{B})}{P(\bar{B})} = \frac{P(A)P(\bar{B})}{P(\bar{B})} = P(A)$, (A) 正确;

由 $P(A | B) > P(A)$ 得 $P(AB) > P(A)P(B)$,

从而 $P(\bar{A} | \bar{B}) = \frac{P(\bar{A}\bar{B})}{P(\bar{B})} = \frac{1 - P(A) - P(B) + P(AB)}{1 - P(B)}$

$$> \frac{1 - P(A) - P(B) + P(A)P(B)}{1 - P(B)} = 1 - P(A) = P(\bar{A}),$$
 (B) 正确;

由 $P(A | B) > P(A | \bar{B})$ 得 $\frac{P(AB)}{P(B)} > \frac{P(A) - P(AB)}{1 - P(B)}$, 整理得 $P(AB) > P(A)P(B)$,

则 $P(A | B) = \frac{P(AB)}{P(B)} > \frac{P(A)P(B)}{P(B)} = P(A)$, (C) 正确, 应选(D).

(9) 【答案】 (B).

【解】 由题知, $\bar{X} \sim N\left(\mu_1, \frac{\sigma_1^2}{n}\right), \bar{Y} \sim N\left(\mu_2, \frac{\sigma_2^2}{n}\right)$,

则 $E(\hat{\theta}) = E(\bar{X}) - E(\bar{Y}) = \mu_1 - \mu_2 = \theta$;

$$\begin{aligned} D(\hat{\theta}) &= D(\bar{X} - \bar{Y}) = D(\bar{X}) + D(\bar{Y}) - 2\text{Cov}(\bar{X}, \bar{Y}) \\ &= \frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{n} - \frac{2}{n} [\text{Cov}(X_1, \bar{Y}) + \text{Cov}(X_2, \bar{Y}) + \cdots + \text{Cov}(X_n, \bar{Y})] \\ &= \frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{n} - \frac{2}{n^2} [\text{Cov}(X_1, Y_1) + \text{Cov}(X_2, Y_2) + \cdots + \text{Cov}(X_n, Y_n)] \\ &= \frac{\sigma_1^2}{n} + \frac{\sigma_2^2}{n} - \frac{2}{n^2} \cdot n\rho\sigma_1\sigma_2 = \frac{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}{n}, \text{应选(B)}. \end{aligned}$$

(10) 【答案】 (A).

【解】 似然函数为

$$L(\theta) = P\{X=1\}P\{X=3\}\cdots P\{X=2\} = \frac{(1-\theta)^3}{8} \cdot \frac{(1+\theta)^5}{4^5},$$

取对数 $\ln L(\theta) = 3\ln(1-\theta) - \ln 8 + 5\ln(1+\theta) - 5\ln 4$,

由 $\frac{d}{d\theta} \ln L(\theta) = \frac{3}{\theta-1} + \frac{5}{\theta+1} = 0$ 得参数 θ 的最大似然估计值为 $\hat{\theta} = \frac{1}{4}$, 应选(A).

二、填空题

(11) 【答案】 $\frac{\sin e^{-1}}{2e}$.

【解】 由 $\frac{dy}{dx} = -\sin e^{-\sqrt{x}} \cdot e^{-\sqrt{x}} \cdot \left(-\frac{1}{2\sqrt{x}}\right) = \frac{e^{-\sqrt{x}} \sin e^{-\sqrt{x}}}{2\sqrt{x}}$ 得 $\frac{dy}{dx} \Big|_{x=1} = \frac{\sin e^{-1}}{2e}$.

(12) 【答案】 6.

【解】 $\int_{\sqrt{5}}^5 \frac{x}{\sqrt{|x^2-9|}} dx = \int_{\sqrt{5}}^3 \frac{x}{\sqrt{9-x^2}} dx + \int_3^5 \frac{x}{\sqrt{x^2-9}} dx$,

而 $\int_{\sqrt{5}}^3 \frac{x}{\sqrt{9-x^2}} dx = -\sqrt{9-x^2} \Big|_{\sqrt{5}}^3 = 2$, $\int_3^5 \frac{x}{\sqrt{x^2-9}} dx = \sqrt{x^2-9} \Big|_3^5 = 4$,

故 $\int_{\sqrt{5}}^5 \frac{x}{\sqrt{|x^2-9|}} dx = 6$.

(13) 【答案】 $\frac{\pi}{4}$.

【解】 所求体积为

$$\begin{aligned} V &= \pi \int_0^1 (\sqrt{x} \sin \pi x)^2 dx = \pi \int_0^1 x \sin^2 \pi x dx = \frac{1}{\pi} \int_0^\pi t \sin^2 t dt \\ &= \frac{1}{\pi} \cdot \frac{\pi}{2} \int_0^\pi \sin^2 t dt = \frac{1}{2} \cdot 2 \int_0^{\frac{\pi}{2}} \sin^2 t dt = \frac{\pi}{4}. \end{aligned}$$

(14) 【答案】 $C + \frac{1}{2}(t^2 - t)$ (C 为任意常数).

【解】 差分方程 $\Delta y_t = t$ 即 $y_{t+1} - y_t = t$,

因为 $a = -1$, 所以一阶齐次差分方程 $y_{t+1} - y_t = 0$ 的通解为 $y_t = C$ (C 为任意常数);

令 $y_{t+1} - y_t = t$ 的特解为 $y_t^* = at^2 + bt$, 代入得

$$a(t+1)^2 + b(t+1) - at^2 - bt = t,$$

则 $2a = 1, a + b = 0$, 解得 $a = \frac{1}{2}, b = -\frac{1}{2}$,

故差分方程的通解为 $y_t = C + \frac{1}{2}(t^2 - t)$ (C 为任意常数).

(15) 【答案】 -5.

$$\begin{aligned} \text{【解】 } f(x) &= \begin{vmatrix} x & x & 1 & 0 \\ 1 & x & 2 & -3 \\ 2 & 1 & x & -3 \\ 2 & -1 & 1 & x-4 \end{vmatrix} = \begin{vmatrix} 0 & 0 & 1 & 0 \\ 1-x & -x & 2 & -3 \\ 1 & 1-x^2 & x & -3 \\ 3 & -1-x & 1 & x-4 \end{vmatrix} \\ &= \begin{vmatrix} 1-x & -x & -3 \\ 1 & 1-x^2 & -3 \\ 3 & -1-x & x-4 \end{vmatrix} = \begin{vmatrix} 1 & -x & 0 \\ x^2 & 1-x^2 & 3x^2-3 \\ 4+x & -1-x & 4x+8 \end{vmatrix} \\ &= (1-x^2)(4x+8) + (1+x)(3x^2-3) + \\ &\quad x[x^2(4x+8) - (x+4)(3x^2-3)], \end{aligned}$$

整理得 x^3 项的系数为 -5.

(16) 【答案】 $\frac{1}{5}$.

【解】 (X, Y) 的可能取值为 $(0, 0), (0, 1), (1, 0), (1, 1)$,

$$P\{X=0, Y=0\} = \frac{1}{2} \times \frac{3}{5} = \frac{3}{10},$$

$$P\{X=0, Y=1\} = \frac{1}{2} \times \frac{2}{5} = \frac{1}{5},$$

$$P\{X=1, Y=0\} = \frac{1}{2} \times \frac{2}{5} = \frac{1}{5},$$

$$P\{X=1, Y=1\} = \frac{1}{2} \times \frac{3}{5} = \frac{3}{10},$$

$$\text{由 } X \sim \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} \text{ 得 } E(X) = \frac{1}{2}, E(X^2) = \frac{1}{2}, D(X) = \frac{1}{4};$$

$$\text{由 } Y \sim \begin{pmatrix} 0 & 1 \\ \frac{1}{2} & \frac{1}{2} \end{pmatrix} \text{ 得 } E(Y) = \frac{1}{2}, E(Y^2) = \frac{1}{2}, D(Y) = \frac{1}{4};$$

$$\text{由 } XY \sim \begin{pmatrix} 0 & 1 \\ \frac{7}{10} & \frac{3}{10} \end{pmatrix} \text{ 得 } E(XY) = \frac{3}{10}, \text{Cov}(X, Y) = E(XY) - E(X)E(Y) = \frac{3}{10} - \frac{1}{4} = \frac{1}{20},$$

$$\text{则 } \rho_{XY} = \frac{\frac{1}{20}}{\frac{1}{2} \times \frac{1}{2}} = \frac{1}{5}.$$

三、解答题

(17) 【解】 $\lim_{x \rightarrow 0^-} \left[a \arctan \frac{1}{x} + (1 + |x|)^{\frac{1}{x}} \right] = -\frac{\pi}{2}a + \lim_{x \rightarrow 0^-} (1-x)^{\frac{1}{x}} = -\frac{\pi}{2}a + e^{-1},$

$$\lim_{x \rightarrow 0^+} \left[a \arctan \frac{1}{x} + (1 + |x|)^{\frac{1}{x}} \right] = \frac{\pi}{2}a + \lim_{x \rightarrow 0^+} (1+x)^{\frac{1}{x}} = \frac{\pi}{2}a + e,$$

因为 $\lim_{x \rightarrow 0} \left[a \arctan \frac{1}{x} + (1 + |x|)^{\frac{1}{x}} \right]$ 存在,

所以 $\frac{\pi}{2}a + e = -\frac{\pi}{2}a + e^{-1}$, 故 $a = \frac{e^{-1} - e}{\pi}$.

(18) 【解】 函数 $f(x, y)$ 的定义域为 $D = \{(x, y) \mid x \neq 0\}$,

由 $\begin{cases} \frac{\partial f}{\partial x} = \frac{2}{x} + \frac{x-1-y^2}{x^3} = 0, \\ \frac{\partial f}{\partial y} = \frac{y}{x^2} = 0 \end{cases}$ 得 $\begin{cases} x = -1, \\ y = 0 \end{cases}$ 或 $\begin{cases} x = \frac{1}{2}, \\ y = 0. \end{cases}$

$$A = \frac{\partial^2 f}{\partial x^2} = -\frac{2}{x^2} + \frac{-2x+3+3y^2}{x^4}, B = \frac{\partial^2 f}{\partial x \partial y} = \frac{-2y}{x^3}, C = \frac{\partial^2 f}{\partial y^2} = \frac{1}{x^2},$$

当 $\begin{cases} x = -1, \\ y = 0 \end{cases}$ 时, $A = 3, B = 0, C = 1$,

因为 $AC - B^2 > 0$ 且 $A > 0$, 所以 $\begin{cases} x = -1, \\ y = 0 \end{cases}$ 为极小值点, 极小值为 $f(-1, 0) = 2$;

当 $\begin{cases} x = \frac{1}{2}, \\ y = 0 \end{cases}$ 时, $A = 24, B = 0, C = 4$,

因为 $AC - B^2 > 0$ 且 $A > 0$, 所以 $\begin{cases} x = \frac{1}{2}, \\ y = 0 \end{cases}$ 为极小值点, 极小值为 $f\left(\frac{1}{2}, 0\right) = -2\ln 2 + \frac{1}{2}$.

(19) 【解】 令 $\begin{cases} x = r \cos \theta, \\ y = r \sin \theta \end{cases} \left(0 \leq \theta \leq \frac{\pi}{4}, 0 \leq r \leq 1 \right)$, 则

$$\begin{aligned} \iint_D e^{(x+y)^2} (x^2 - y^2) dx dy &= \int_0^{\frac{\pi}{4}} d\theta \int_0^1 r e^{r^2} \cdot e^{r^2 \sin 2\theta} \cdot r^2 \cos 2\theta dr \\ &= \frac{1}{2} \int_0^{\frac{\pi}{4}} d\theta \int_0^1 e^{r^2} \cdot e^{r^2 \sin 2\theta} \cdot r^2 \cos 2\theta d(r^2) \\ &= \frac{1}{2} \int_0^{\frac{\pi}{4}} d\theta \int_0^1 e^t \cdot e^{t \sin 2\theta} \cdot t \cos 2\theta dt \\ &= \frac{1}{4} \int_0^1 e^t dt \int_0^{\frac{\pi}{4}} e^{t \sin 2\theta} d(t \sin 2\theta) = \frac{1}{4} \int_0^1 e^t \cdot \left(e^{t \sin 2\theta} \Big|_0^{\frac{\pi}{4}} \right) dt \\ &= \frac{1}{4} \int_0^1 e^t \cdot (e^t - 1) dt = \frac{1}{4} \left(\frac{1}{2} e^{2t} \Big|_0^1 - e^t \Big|_0^1 \right) \\ &= \frac{1}{4} \left(\frac{e^2}{2} - e + \frac{1}{2} \right) = \frac{(e-1)^2}{8}. \end{aligned}$$

(20) 【解】 (I) 由 $xy' - (n+1)y = 0$ 得 $y' - \frac{n+1}{x}y = 0$,

通解为 $y_n(x) = Ce^{-\int \frac{n+1}{x} dx} = Cx^{n+1}$,

由 $y_n(1) = \frac{1}{n(n+1)}$ 得 $C = \frac{1}{n(n+1)}$, 故 $y_n(x) = \frac{x^{n+1}}{n(n+1)}$.

(II) 对于级数 $\sum_{n=1}^{\infty} \frac{x^{n+1}}{n(n+1)}$, 由 $\lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = 1$ 得收敛半径 $R = 1$,

当 $x = \pm 1$ 时, 级数 $\sum_{n=1}^{\infty} \left| \frac{(\pm 1)^{n+1}}{n(n+1)} \right| = \sum_{n=1}^{\infty} \frac{1}{n(n+1)} = 1$, 故收敛域为 $[-1, 1]$.

令 $S(x) = \sum_{n=1}^{\infty} \frac{x^{n+1}}{n(n+1)}$, 则

$$\begin{aligned} S(x) &= \sum_{n=1}^{\infty} \frac{x^{n+1}}{n(n+1)} = \sum_{n=1}^{\infty} \frac{x^{n+1}}{n} - \sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1} = x \sum_{n=1}^{\infty} \frac{x^n}{n} - \sum_{n=1}^{\infty} \frac{x^n}{n} + x \\ &= (1-x) \ln(1-x) + x \quad (-1 \leq x < 1), \end{aligned}$$

故 $S(x) = \begin{cases} 1, & x = 1, \\ (1-x) \ln(1-x) + x, & -1 \leq x < 1. \end{cases}$

(21) 【解】 由 $|\lambda E - A| = \begin{vmatrix} \lambda - 2 & -1 & 0 \\ -1 & \lambda - 2 & 0 \\ -1 & -a & \lambda - b \end{vmatrix} = (\lambda^2 - 4\lambda + 3)(\lambda - b) = 0$ 得

矩阵的特征值为 $\lambda_1 = 1, \lambda_2 = 3, \lambda_3 = b$.

情形一: $b = 1$

因为 A 相似于对角矩阵, 所以 $r(E - A) = 1$,

而 $E - A = \begin{pmatrix} -1 & -1 & 0 \\ -1 & -1 & 0 \\ -1 & -a & 0 \end{pmatrix} \rightarrow \begin{pmatrix} -1 & -1 & 0 \\ 0 & a-1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, 故 $a = 1$,

由 $E - A \rightarrow \begin{pmatrix} 1 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ 得 $\lambda = 1$ 的线性无关的特征向量为 $\alpha_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}, \alpha_2 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$;

由 $3E - A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ -1 & -1 & 2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$ 得 $\lambda = 3$ 的特征向量为 $\alpha_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$,

令 $P = \begin{pmatrix} -1 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix}$, 则 $P^{-1}AP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.

情形二: $b = 3$

因为 A 相似于对角矩阵, 所以 $r(3E - A) = 1$,

而 $3E - A = \begin{pmatrix} 1 & -1 & 0 \\ -1 & 1 & 0 \\ -1 & -a & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & -1 & 0 \\ 0 & -a-1 & 0 \\ 0 & 0 & 0 \end{pmatrix}$, 故 $a = -1$.

由 $E - A = \begin{pmatrix} -1 & -1 & 0 \\ -1 & -1 & 0 \\ -1 & 1 & -2 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$ 得 $\lambda = 1$ 的特征向量为 $\alpha_1 = \begin{pmatrix} -1 \\ 1 \\ 1 \end{pmatrix}$;

由 $3E - A \rightarrow \begin{pmatrix} 1 & -1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ 得 $\lambda = 3$ 的线性无关的特征向量为 $\alpha_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \alpha_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$,

令 $P = \begin{pmatrix} -1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$, 则 $P^{-1}AP = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$.

(22) 【解】 (I) X 的密度函数为

$$f_X(x) = \begin{cases} 1, & 0 < x < 1, \\ 0, & \text{其他.} \end{cases}$$

(II) 由 $Y = 2 - X$ 得 $Z = \frac{2 - X}{X}$,

故 $F_Z(z) = P\{Z \leq z\} = P\left\{\frac{2}{X} - 1 \leq z\right\}$,

当 $z < 1$ 时, $F_Z(z) = 0$;

当 $z \geq 1$ 时, $F_Z(z) = P\left\{X \geq \frac{2}{z+1}\right\} = \int_{\frac{2}{z+1}}^1 1 dx = 1 - \frac{2}{z+1} = \frac{z-1}{z+1}$,

即 $F_Z(z) = \begin{cases} 0, & z < 1, \\ \frac{z-1}{z+1}, & z \geq 1, \end{cases}$

故 Z 的密度函数为 $f_Z(z) = \begin{cases} 0, & z < 1, \\ \frac{2}{(z+1)^2}, & z \geq 1. \end{cases}$

(III) $E\left(\frac{X}{Y}\right) = E\left(\frac{X}{2-X}\right) = \int_0^1 \frac{x}{2-x} dx = 2\ln 2 - 1$.